

A Measure-Theoretic Discrete Curvature Framework for Metabolic Gene Sensitivity: From Active-Set Geometry to Transcriptional Response

Z.ai

September 2026

Abstract

Parametric flux balance analysis defines, for every metabolic network, a piecewise-affine map from environmental and genetic parameters to optimal fluxes. We show that the distributional second derivative of this map is a matrix-valued Radon measure concentrated on the codimension-one strata of the active-set complex, and that this measure — not a smooth curvature — is the canonical carrier of the network’s rerouting geometry: loop holonomy scales linearly in loop size (slope 1.00, versus the quadratic scaling of any smooth map), and the second-order response mass concentrates (100.0000%) on operational active-set switches. We prove a refinement–resolution bridge (Theorem B’): under mesh refinement the atomic measure converges weakly to the smooth curvature density, the dual-cell reconstruction converges strongly in L^1 , and the total-variation claims of earlier formulations are replaced by an exact counterexample calculus; a resolution parameter σ separates the discrete and smooth regimes through the measured window $h \ll \sigma \ll L_{\text{var}}$. From the measure we derive a gene-level sensitivity metric κ^μ (measure mass along a parametric trajectory) and show, on *E. coli* (M3D microarrays, $n = 424$ genes), that κ^μ predicts carbon-depletion transcriptional response ($r = +0.395$, $p = 2.6 \times 10^{-17}$; partial $r = +0.269$ controlling reference level; metric-invariant $\rho = 0.99998$ against the lexicographic definition), while the association does not propagate to the protein layer — consistent with a model in which cells transcribe the potential for rerouting while buffering its translation. Double-knockout epistasis aligns with active-set structure ($\rho_S = 0.865$), and genotype loops close with phenotypic memory (66% non-reverting). The framework unifies three previously distinct sensitivity objects as one measure at different resolutions.

Contents

1	Introduction	3
2	Notation protocol and the R/D/E hierarchy	3
3	The discrete curvature measure of parametric FBA	3
3.1	[D1] The measure	3
3.2	[D2] The measure-theoretic sensitivity metric	4
3.3	[D3] The value-function layer (Alexandrov)	4
3.4	[D4] The value–flux coupling	5
4	The refinement–resolution bridge (Theorem B’)	6
4.1	[R-port] Bridge to parametric FBA	8
5	Computational validation	8
5.1	[E-M1] Active-set curvature verification	8
5.2	[E-M3] Double-knockout epistasis and path dependence	8

5.3	[E-M4] Regime dial	8
5.4	[E-V1] The value function as canonical carrier	9
6	Empirical association and protein-layer buffering	9
6.1	[E-E22] The gene panel and GPR classes	9
6.2	[E-E23] Multi-condition support	9
6.3	[E-V5] Transcriptional response under carbon depletion	9
6.4	[E-V6] The layer decision	10
6.5	[E-V7] Path robustness: what survives a trajectory change	10
6.6	[E-E25] Platform-by-class disambiguation (2×2)	11
6.7	[E-E26] Protein abundance versus protein change	11
6.8	[E-E27] Protein-layer null	11
7	Discussion: one measure, several resolutions	11
8	Methods	12
A	Layer-1 porting map	12

1 Introduction

The object. Let a metabolic network with m reactions be operated by flux balance analysis with parameters $\theta \in \Theta \subset \mathbb{R}^p$ (uptake bounds, knockdown scalings) entering the constraints affinely. The optimal flux map $v : \Theta \rightarrow \mathbb{R}^m$ is continuous and piecewise affine (parametric linear programming), with breakpoints on the active-set strata of the constraint complex. Its distributional second derivative $\mu = D^2v$ is therefore not a function but a *measure* — the discrete curvature carrier of the network’s rerouting geometry.

The claim structure. This manuscript establishes, in order:

- (i) **[R1]** the atom structure and weak convergence of μ_h under refinement (Theorem B’, §4), with the exact counterexample calculus that separates what is provable from what is false in earlier formulations;
- (ii) **[R2]** the two-layer separation: the codimension-one Hessian layer is the unbiased curvature estimator; the codimension-two angle-defect layer converges to the Gauss-map area density with an exact sec-law bias (Proposition 4.3);
- (iii) **[R3]** the regime dichotomy: piecewise-affine maps with fixed kink complexes produce $O(\varepsilon)$ loop holonomy (measured slope 1.00) where smooth maps produce $O(\varepsilon^2)$;
- (iv) **[R4]** the value function $\Phi = c_{\text{bio}}^\top v^*$ is the canonical tie-break-free carrier, with shadow-price jumps measurable to 6–7 digits against Danskin envelopes;
- (v) **[E-series]** the measure-derived metric κ^μ predicts carbon-depletion transcriptional response ($r = +0.395$), generalizes to oxygen-limitation and carbon-switch trajectories with matched responses ($r = +0.32, +0.22$; E-V7), and is buffered at the protein layer (§6).

Positioning. Kochanowski et al. demonstrated that global metabolic coordination under catabolic limitation is largely passive at the protein/metabolite level, driven by Crp and local metabolite concentrations. Our work asks a different question: whether a gene-specific, network-derived rerouting necessity predicts transcriptional response. We find that it does, and that this transcript-level association does not propagate to the protein layer — consistent with a model in which cells transcribe the potential for rerouting while buffering its translation.

2 Notation protocol and the R/D/E hierarchy

- **[R- n]** blocks are proved results (theorems, propositions, lemmas) or machine-verified claims with reproducible artifacts.
- **[D- n]** blocks are definitions, constructions, and design decisions (metric definitions, engine specifications).
- **[E- n]** blocks are executed experiments with deposited artifacts (data files, figures, statistics).
- Conjectures are labelled as such; plausible-but-unproven claims are demoted, never promoted.

3 The discrete curvature measure of parametric FBA

3.1 [D1] The measure

Definition 3.1 (Atomic curvature measure). Let $u : \Omega \rightarrow \mathbb{R}^m$ be continuous and piecewise affine on a polyhedral complex $\mathcal{C}$ covering Ω . For each codimension-one facet F of $\mathcal{C}$ shared by cells τ^- ,

τ^+ , let $[\nabla u]_F = \nabla u|_{\tau^+} - \nabla u|_{\tau^-}$, let n_F be the unit conormal oriented $\tau^- \rightarrow \tau^+$, and let $|F|$ be the facet measure. The *atomic curvature measure* is the matrix-valued Radon measure

$$\mu = D^2 u = \sum_F A_F \delta_F, \quad A_F = [\nabla u]_F \otimes n_F |F| \in \mathbb{R}^{m \times p \times p}.$$

Remark 3.2 (Support). μ is concentrated on the $(p-1)$ -skeleton. The $(p-2)$ -skeleton carries a different measure (the angle-defect layer, Proposition 4.3); the two must not be conflated. In $p=1$ the atoms are the slope jumps at breakpoints.

3.2 [D2] The measure-theoretic sensitivity metric

Definition 3.3 (κ^μ). For a C^1 parameter trajectory $\theta(t)$, $t \in [0, T]$, and its piecewise-affine optimal-flux response $v(\theta(t))$, let $\{t_e\}_e$ be the event times at which the trajectory crosses active-set strata, with atoms $A_e = A_{F(e)}$ evaluated on the crossed facet. Define

$$\kappa^\mu(g) = \frac{1}{T} \sum_e \|A_e\| w_e(g),$$

where $\|\cdot\|$ is the entrywise 1-norm on $\mathbb{R}^{m \times p \times p}$ and $w_e(g) \in \{0, 1\}$ selects the events whose crossed stratum changes the activity of reactions assigned to gene g (max over its reactions, per the GPR association).

Remark 3.4 (κ^μ is the safe regime of the reconstruction). κ^μ integrates measure mass along a one-dimensional trajectory — the telescoping regime in which total variation and mass are automatically consistent (R2 remarks in §4); its resolution independence (measured mass 288.77 stable under $4 \times 8 \times$ trajectory refinement) is exactly the L^1 -reconstruction regime of Theorem B'(iv). Its per-reaction masses are the components of the flux-layer crease measure whose objective contraction is $D^2\Phi$ (Theorem 3.11): the metric is the primal layer of the canonical object.

Remark 3.5 (Locked status, tie-breaking, and layer assignment). The definition of κ^μ is final for the empirical claims of this manuscript. (i) The carrier is the flux-layer signed crease measure $D^2 v^*$ (Definition 3.1 applied to $u = v^*$) of the *three-stage lexicographic* optimal flux (§8); the lexicographic tie-break (biomass, then ℓ_1 flux, then seeded deterministic weights) is part of the metric's definition, not an implementation detail — the flux layer is selection-dependent by construction, and the tie-break is declared rather than eliminated. (ii) The per-reaction mass is the unsigned total variation $\frac{1}{T} \sum_e \|A_e\|_1$ along the parameter trajectory. (iii) The value layer $D^2\Phi$ enters only as a *derived diagnostic* (Theorem 3.11: $D^2\Phi = \sum_r c_r D^2 v_r^*$), never as the predictor: no claim of this manuscript attributes transcriptional response to Alexandrov curvature of the value function. (iv) The layer correlated with transcriptomics is the codimension-one Hessian/crease layer of the flux map — not the Monge–Ampère atom layer (Proposition 3.13), which requires concavity and serves as the dual-face diagnostic. Robustness to the declared selection is measured, not assumed: the within-layer rank identity $\rho(\kappa^\mu, \kappa_V^{\text{lex}}) = 0.99998$ (E-V5), and the trajectory substitutions of E-V7 reproduce the association under the same declared selection on two further paths.

3.3 [D3] The value-function layer (Alexandrov)

Proposition 3.6 (Existence and tie-break-freeness). *Let $\Phi(\theta) = \max\{c^\top v : Sv = 0, \ell(\theta) \leq v \leq u(\theta)\}$ with ℓ, u affine, feasible and bounded. Then Φ is concave (pointwise infimum of affine dual objectives over a θ -independent dual feasible set), and $-\Phi$ carries a symmetric matrix-valued Radon Hessian measure $D^2(-\Phi)$ (Alexandrov), independent of the optimal-flux selection rule. Concavity fails in general under OR (isoenzyme) GPR rules for simultaneous multi-gene capacity vectors; single-gene axes, exchange-bound families, and AND-only subnetworks retain it. For OR-containing rules the value function remains continuous piecewise affine and carries the signed crease measure of Proposition 3.12; only the Monge–Ampère layer requires the concave class.*

Proposition 3.7 (Two-layer structure of $D^2(-\Phi)$). *Let f be convex piecewise affine. (i) The singular part of D^2f is supported on the codimension-one facets, with density $[\partial_n f](g^+ - g^-) \otimes n$ per unit facet measure; it has no atoms: the mass in an ε -neighborhood of a codimension-two vertex is $O(\varepsilon)$ (measured log-log slope 1.000). (ii) The Monge–Ampère measure $\det D^2f$ is atomic at codimension-two vertices, each atom equal to the volume of the normal fan $|\text{conv}\{\nabla f(\tau) : \tau \ni \text{vertex}\}|$ (measured: constant 0.5 under ε -shrinking).*

Proposition 3.8 (Dual optimal face identity). *At a codimension-two vertex θ_v of a parametric-LP value function, the Monge–Ampère atom of $-\Phi$ equals the area of the normal fan, and the fan is the image of the LP dual optimal face under $\theta \mapsto U^\top y$:*

$$\det D^2(-\Phi)(\{\theta_v\}) = |\text{conv}\{\nabla C_i\}| = |\{U^\top y : (\lambda, y) \text{ optimal dual}\}|.$$

Machine-verified on a two-parameter max-flow LP (the phenotype phase plane in miniature): vertex refined to machine precision by exact triple-tie solve, $\Phi(\theta_v)$ matching the LP value to 10^{-15} , fan area 0.235 vs dual-face enumeration area 0.235000068 (16 direction LPs over the dual optimal face).

Remark 3.9 (Layer separation and PhPP positioning). The phenotype phase planes of Edwards–Ibarra–Palsson are the chamber complex of Φ ; their sector corners are the atoms of the Monge–Ampère layer. The empirical κ^μ of Definition 3.3 is a functional of the *flux-map* measure D^2v (Definition 3.1), which is tie-break-sensitive: in a controlled follower-variable LP, the value function stays affine and identical under two lexicographic tie-breaks while the follower’s second-difference mass flips $0.0100 \leftrightarrow 10^{-13}$. The controlled relation between the value layer and the flux layer is now a theorem (Theorem 3.11 and Corollary 4.9); the association results of §6 stand on the flux layer alone, which is where the evidence (E-V5, E-V6) actually lives.

3.4 [D4] The value–flux coupling

Lemma 3.10 (Lex-selection regularity). *Let ℓ, u be continuous piecewise affine and let $v^*(\theta)$ be the three-stage lexicographic optimal flux (§8). Then v^* is well defined, continuous, and piecewise affine; its graph over each GPR cell is polyhedral in (v, θ) -space (each lexicographic stage adjoins the graph of a piecewise-affine value function, which is polyhedral, and an injective polyhedral projection is piecewise affine). Measured: path segment residuals $\leq 8 \times 10^{-14}$ (E-M1); Φ piecewise affine at 4.2×10^{-13} (E-V1).*

Theorem 3.11 (Value–flux crease coupling). $\Phi = c^\top v^*$ pointwise; hence, as symmetric matrix-valued signed Radon measures on the parameter domain,

$$D^2\Phi = \sum_r c_r D^2v_r^*.$$

Along any piecewise-affine trajectory, at every crease time t_k : $\Delta\Phi'(t_k) = c^\top \Delta v'(t_k)$, and $|\Delta\Phi'| \leq \|c\|_\infty \|\Delta v'\|_1$.

- (i) (**Visibility dichotomy.**) *An event with unique optima on both sides is a transversal optimal-vertex switch and is necessarily objective-moving; objective-invisible events ($c^\top \Delta v' = 0$, $\Delta v' \neq 0$) are exactly the degenerate reroutings inside ≥ 1 -dimensional optimal faces (mask-type events). The tie-break sensitivity of the flux layer is concentrated in the invisible events; the c -contraction is tie-break-free because it annihilates the degenerate layer.*
- (ii) (**Sparse-objective corollary.**) *If $c = \gamma e_{\text{bio}}$, then $D^2\Phi = \gamma D^2v_{\text{bio}}^*$: the value layer is the crease measure of the single biomass component, and per-gene attribution from the value layer is structurally impossible.*

Machine-verified: identity error 0.0 at all 12 events and all 4 clusters of the iML1515 cut ($\Phi = c^\top v^$ to 10^{-9}); 11 of 12 events objective-invisible, with flux ℓ_1 slope jumps up to 1.6×10^4 against $|c^\top \Delta v'| \leq 1.5 \times 10^{-7}$; 150 random dense-objective LPs to 1.1×10^{-13} (5 events, all objective-moving — the dichotomy); the degenerate follower family 60/60 with invisible kinks at identity error 2.0×10^{-13} ; two-dimensional mixed second differences at crease-straddling boxes to 6.4×10^{-16} .*

Proposition 3.12 (Semiconvexity collapse; signed layer for mixed GPRs). *A continuous piecewise-affine function is semiconvex (resp. semiconcave) iff it is convex (resp. concave) — semiconvexity is not an available weakening for LP value functions (measured: the required constant blows up like $1/(2h)$; $\lambda \cdot h_{\max} \equiv 0.500$ across $\lambda = 1 \dots 10^{12}$). Nevertheless every continuous piecewise-affine Φ , for any GPR structure, carries a finite signed symmetric Radon Hessian measure on its crease skeleton with per-crease sign flags: for the OR-plus-cap value function $\min(\max(\theta_1, \theta_2), K)$, the diagonal crease density is positive semidefinite (eigenvalues $\{0, \sqrt{2}\}$) and the cap crease negative semidefinite ($\{-1, 0\}$). Concavity is required only by the Monge–Ampère layer and the dual-face identity; the signed Hessian layer applies to all genes. Under multiplicative activity scaling with $\ell \leq 0 \leq u$, $\Phi = \Psi(a(\theta))$ with Ψ concave nondecreasing: AND-only (min-tree) GPRs retain concavity, top-level OR nodes can break it, and disaggregating isozymes into separate reactions restores affine bounds while replacing max-semantics by sum-semantics (capacity 1 vs 2 at full expression) — a different biological model, not a regularization.*

Proposition 3.13 (Monge–Ampère atoms are determinants, not products). *For convex piecewise-affine f on $\mathbb{R}^2$ at a generic vertex with incident gradients n_1, n_2, n_3 and codimension-one gradient jumps j_i : the atom equals $\frac{1}{2}|\det(j_1, j_2)| = |\text{conv}\{n_i\}| = \frac{1}{2}|j_1||j_2|\sin \angle$; the product $|j_1||j_2|$ overestimates by $2/\sin \angle \geq 2$, with equality iff the jumps are orthogonal (measured on 400 random max-of-affine vertices: determinant = fan area to 8.9×10^{-16} ; product/atom median 3.296, minimum exactly 2.000; the $2/\sin \angle$ law to 2.5×10^{-12} ; in p dimensions, atom = $(1/p!)|\det(j_1, \dots, j_p)|$ for simplex fans).*

Remark 3.14 (Sign and normalization conventions). Definition 3.1 applied to $u = -\Phi$ is the signed crease measure of the value function: per facet F , the density $A_F = [\nabla(-\Phi)]_F \otimes n_F |F|$, a symmetric matrix whose eigenvalue signs flag the crease’s convexity/concavity (Proposition 3.12). This is the explicit extension of Alexandrov’s setting to arbitrary GPR structure: within the concave class $-\Phi$ is convex, every $A_F \succeq 0$, and the measure coincides with the Alexandrov Hessian measure; outside it, the same definition produces the signed layer. All masses in this manuscript are unsigned total variations (entrywise 1-norm for matrix atoms, trajectory sums for κ^μ); Monge–Ampère atoms are unsigned $\frac{1}{2}|\det(j_1, j_2)|$ equal to the normal-fan volume, normalized to $(1/p!)|\det(j_1, \dots, j_p)|$ for simplex fans in dimension p (Proposition 3.13). Signed determinants appear only in the concave class, where they are nonnegative.

Remark 3.15 (What this closes). Theorem 3.11 is the exact form of the value–flux relation previously left open: the empirical κ^μ is the per-reaction total variation of the flux-layer crease measure whose objective contraction is $D^2\Phi$ — the metric is the primal layer of the canonical object, justified by identity rather than by rank agreement.

4 The refinement–resolution bridge (Theorem B')

Throughout this section $\Omega \subset \mathbb{R}^p$ is compact, $p \geq 1$, $u \in C^2(\Omega; \mathbb{R}^m)$, $\{\mathcal{T}_h\}$ is a shape-regular family of conforming triangulations with mesh size $h = \max_\tau \text{diam}(\tau) \rightarrow 0$, and u_h is the continuous piecewise-affine interpolant of u on $\mathcal{T}_h$.

Theorem 4.1 (Refinement–resolution bridge, corrected form). *Let $\mu_h = D^2 u_h$ (Definition 3.1) and $\mu = D^2 u \text{ dvol}$. Then:*

- (i) **(Atoms / support.)** μ_h is a finite matrix-valued Radon measure concentrated on the $(p-1)$ -skeleton of $\mathcal{T}_h$ with atoms $A_F = [\nabla u_h]_F \otimes n_F |F|$; there are no atoms on lower-dimensional strata.
- (ii) **(Weak convergence.)** $\mu_h \Rightarrow \mu$ weakly as matrix-valued measures: for every $\varphi \in C_c^\infty(\Omega)$, $\langle \mu_h, \varphi \rangle \rightarrow \langle \mu, \varphi \rangle$. Moreover $\|\mu_h - \mu\|_{\text{KR}} \rightarrow 0$ in the Kantorovich–Rubinstein (flat) norm.
- (iii) **(Total variation: the honest statement.)** $\sup_h \|\mu_h\|_{\text{TV}} \leq C|u|_{C^2}|\Omega|$ (under quasi-uniformity) and $\liminf_h \|\mu_h\|_{\text{TV}} \geq \|\mu\|_{\text{TV}}$; but $\|\mu_h\|_{\text{TV}} \rightarrow \|\mu\|_{\text{TV}}$ is false in general — see Remark 4.2. It holds in the one-signed regime (convex or concave data restricted to one-dimensional parameter families), which is the regime of the parametric-LP value function and of κ^μ .
- (iv) **(No-mass-loss, correct form: L^1 reconstruction.)** On structured complexes, the per-cell reconstructed density $\rho_h(\text{cell}) = |\text{cell}|^{-1} \sum_{F \rightarrow \text{cell}} A_F$ converges to $D^2 u$ strongly in L^1 with rate $O(h)$ for $u \in C^3$; consequently $\|\rho_h\|_{L^1} \rightarrow \|\mu\|_{\text{TV}}$. On unstructured complexes the naive three-facet patch is not the dual cell; the general statement requires the discrete-duality (dual-diamond) pairing.
- (v) **(Resolution window.)** For the coarse-grained family $\mu_{h,\sigma} = \mu_h * \varphi_\sigma$: at fixed h , the limit $\sigma \rightarrow 0$ is the atomic measure; the smooth object exists only as a family member at matched resolution through the window $h \ll \sigma \ll L_{\text{var}}$ (the measured regime dial, §5.3).

Proof of (ii) (two lines). For $\varphi \in C_c^\infty(\Omega)$, $\langle D^2 u_h, \varphi \rangle = \int_\Omega u_h D^2 \varphi \, \text{dvol}$ by the definition of distributional derivatives (no boundary terms arise). Since $\|u - u_h\|_\infty \leq Ch^2|u|_{C^2}$ on shape-regular families, $\int u_h D^2 \varphi \rightarrow \int u D^2 \varphi = \langle \mu, \varphi \rangle$. The flat-norm statement follows by density of $\{c\varphi : \text{Lip}(\varphi) \leq 1\}$ in the KR dual cone and the uniform mass bound of (iii). $\square$

Remark 4.2 (Exact counterexample calculus). On the right-triangle mesh of $[0, 1]^2$ with Hessian $H = \begin{pmatrix} a & b \\ b & c \end{pmatrix}$, the per-cell atoms are $A_{\text{diag}} = bh^2 \mathbf{1}\mathbf{1}^\top$, $A_{\text{vert}} = (a-b)h^2 e_x \otimes e_x$, $A_{\text{horiz}} = (c-b)h^2 e_y \otimes e_y$; their sum is exactly $h^2 H$ (machine-verified to 0.0), while the entrywise total variation per cell is $|a-b| + |c-b| + 4|b|$ against the target $|a| + |c| + 2|b|$. Hence for $u = xy$ the ratio is exactly $3 - 1/n$, for the strictly convex $u = 2x^2 - xy + 2y^2$ it is $1.4 - 1/n$ (measured: 2.9922 and 1.3922 at $n = 128$), and for generic smooth maps 3.7–4.4. Total-variation convergence fails generically — including under convexity — while weak convergence holds at rate $O(h^2)$ throughout.

Proposition 4.3 (Two-layer separation and the sec law). *Let $K_h = \sum_v \text{defect}(v) \delta_v$ be the vertex angle-defect measure of the polyhedral graph $\{(x, u_h(x))\}$. Then K_h converges weakly to the Gauss-map area density*

$$\frac{\det D^2 u}{(1 + |\nabla u|^2)^{3/2}} \, \text{dvol} = K \sqrt{1 + |\nabla u|^2} \, \text{dvol}, \quad K = \frac{\det D^2 u}{(1 + |\nabla u|^2)^2},$$

i.e. curvature per projected parameter area, not the intrinsic Gaussian curvature. The bias factor is exactly $\sqrt{1 + |\nabla u|^2}$ (the sec law): machine-verified to 0.0% excess across 13 probes spanning tilts 0.11–2.83 and three tilt directions. Recovering intrinsic curvature requires the graph area correction.

Remark 4.4 (Consequences). (i) The codimension-one Hessian layer is the unbiased estimator and the canonical carrier for the bridge; (ii) Regge/angle-defect formulations need the area correction of Proposition 4.3; (iii) defects of a fixed complex (as measured in the M4b census) are scale-invariant $O(1)$ objects and are not contradicted by Proposition 4.3, which concerns refinement limits.

Proposition 4.5 (Regime dichotomy; M4a in closed form). *Let $B(\varepsilon)$ be the curvature mass captured by a loop of radius ε . For a C^2 map, $B(\varepsilon) = O(\varepsilon^2)$; for a piecewise-affine map whose kink complex meets the loop transversally, $B(\varepsilon) = O(\varepsilon)$. Measured slopes on the M4a design: 2.000 (smooth) and 1.00 (parametric FBA, 76-pair perturbation scan).*

4.1 [R-port] Bridge to parametric FBA

Conjecture 4.6 (mpLP refinement bridge). Let $\{\Phi_n\}$ be a sequence of parametric-LP value functions (minima of tangent-plane families; θ in the right-hand side; nested outer approximations) converging uniformly to a C^2 limit f , with induced complexes $\mathcal{C}_n$. Then $\mu_n = D^2\Phi_n$ converges weakly to D^2f dvol provided the complexes satisfy a per-cell discrete-duality identity (structured complexes) or a stable discrete-duality (dual-diamond) pairing. Along one-dimensional parameter cuts the conclusion holds unconditionally by monotone telescoping of the concave restriction.

Remark 4.7 (Status). The one-dimensional cut case is machine-verified (independent-seed reproduction of the refinement experiment: folded Wasserstein distance $0.058 \rightarrow 0.0039$, mass ratio $0.939 \rightarrow 1.0009$ across $n = 4 \rightarrow 128$). The general complex case is open; the discrete-duality pairing stability is the identified missing assumption.

Remark 4.8 (Existence vs. resolution). Proposition 3.6 supplies existence of the curvature measure without refinement sequences, smoothing, or interpolation; Theorem B' governs what a discretely sampled trajectory can *resolve* of that measure, and at what rate. The two layers are complementary: the Alexandrov layer removes the extrinsic hypotheses, the refinement bridge keeps the quantitative rate.

Corollary 4.9 (Decoupling; resolves the value–flux layer relation). *Theorem 3.11 resolves the layer relation exactly: value events are a subset of flux events, never the converse, and the flux events invisible to the value layer are the degenerate reroutings (mask-type events — rerouting without growth loss, the RC6/Kochanowski pattern). On the iML1515 cut: 12 flux events vs 1 value atom, 11 objective-invisible; on the E24 trajectory (E-V6) the value layer carries no network events at all. The association of κ^μ with transcriptional response is therefore a flux-layer property by structural necessity, not merely by empirical choice.*

5 Computational validation

5.1 [E-M1] Active-set curvature verification

Twelve parameter sweeps (glucose, oxygen, eight gene knockdowns) on *E. coli* iML1515 with an iJO1366 replication, solved by a three-stage lexicographic pFBA engine (deterministic optimum; fixed seeded tie-break weights after the discovery of pFBA vertex degeneracy). Result: 100.0000% of second-order response mass concentrates on operational active-set switches in every sweep that crosses critical-region boundaries (mass 0.934–1.0; AUC 0.83–1.00; MWU $p \leq 10^{-4}$); paths are piecewise affine to machine precision between events (relative segment residuals $\leq 8 \times 10^{-14}$); negative controls behave as theory dictates (unused-pathway knockdown: zero events, zero curvature; single-critical-region sweep: noise-level $D^2 \sim 10^{-11}$).

5.2 [E-M3] Double-knockout epistasis and path dependence

1,516 single knockouts and 2,779 double-knockout pairs across five panels. Epistasis $\varepsilon_{ij} = \kappa_{ij} - \kappa_i - \kappa_j$: all 40 synthetic-lethal pairs are isozyme redundancies (tktA/tktB, acnA/acnB, metE/metH, ...) with pure-emergence epistasis; $|\varepsilon_{ij}|$ aligns with active-set footprint overlap (Spearman $\rho_S = 0.865$, $J_{\text{support}} = 0.800$); sequential (L_1 -MOMA) knockouts open nonzero commutators in 25% of active pairs; closed genotype loops fail to return to the initial state in 66% of pairs (phenotypic memory; median holonomy ~ 110).

5.3 [E-M4] Regime dial

The smoothing identity $D^2(v * \phi_\sigma) = (D^2v) * \phi_\sigma$ holds to kernel self-test error 1.2×10^{-6} ; measured crossover $\varepsilon^*/\sigma \in [2.45, 4.11]$ (median 3.1) separates the $O(\varepsilon^2)$ smooth regime from the $O(\varepsilon)$

discrete regime. The *M4a* scaling slope is 1.00 on the interacting stratum; the *M4b* census resolves the thin-wedge sliver structure (net measure jump 8.9 vs. nominal ± 1884.6).

5.4 [E-V1] The value function as canonical carrier

$\Phi = c_{\text{bio}}^\top v^*$ is single-valued with no tie-breaking; it is piecewise affine at 4.2×10^{-13} on the flux-event partition, carries one real atom $\Delta\Phi' = -0.006439$ at $t = 0.0358286$, and decouples from the flux-jump hierarchy (value/flux mass ratio 1.7×10^{-6}). Danskin: finite-difference shadow prices match envelope marginals to 6–7 digits. A controlled follower-variable LP reproduces the same structure: the value function stays affine and identical under two lexicographic tie-breaks while the follower’s second-difference mass flips $0.0100 \leftrightarrow 10^{-13}$ — the value layer is canonical (Proposition 3.6), the flux-strain layer is not (Corollary 4.9). The coupling identity of Theorem 3.11 is machine-verified on this same cut: $\Delta\Phi' = c^\top \Delta v'$ with error 0.0 at all 12 events and all 4 clusters, $\Phi = c^\top v^*$ to 10^{-9} , and the 11 invisible events carry flux ℓ_1 slope jumps up to 1.6×10^4 against $|c^\top \Delta v'| \leq 1.5 \times 10^{-7}$ — the decoupling is the objective contraction, not a loss of network information.

6 Empirical association and protein-layer buffering

6.1 [E-E22] The gene panel and GPR classes

Reaction-based sampling on the E22 physiology (iJO1366): 438 of 2,583 reactions active (17.0%), 435 genes with at least one active reaction. GPR complexity classes over active reactions: single-gene 254, isozyme-OR 96, complex-AND 36, mixed 18, no-GPR 34 — the classes that drive the concavity analysis of Proposition 3.12. Panel construction is consistent with the precursor metric family to the digit (per-gene values and panel-level correlations reproduced exactly); the precursor geometric metric κ_V is weak at the panel level (r from -0.063 unmasked to $+0.084$ masked) — the discrepancy that motivated the recalibration to κ^μ (E-V5).

6.2 [E-E23] Multi-condition support

Nine FBA conditions (baseline carbon decline; galactose, trehalose, glycerol-3-P, and proline-N carbon sources; anaerobic-nitrate, oxidative, and osmotic designs): active-reaction support 438–454 per condition, union 537 (20.8%); genes with nonzero κ_V expand $435 \rightarrow 525$ across conditions. The endogeneity audit is clean (feed caps non-binding except the designed glycerol-3-P limitation; imposed-burden magnitudes flagged as not interpretable). The rerouting metric is well posed across carbon sources and stress designs: its support *expands* with condition diversity rather than migrating.

6.3 [E-V5] Transcriptional response under carbon depletion

On the deterministic lexicographic trajectory (E22 physiology, iJO1366), κ^μ (Definition 3.3) versus log-fold carbon-depletion response (M3D, $n = 424$ genes): Pearson $r = +0.395$ ($p = 2.6 \times 10^{-17}$), Spearman $\rho_S = +0.414$, partial $r = +0.269$ controlling reference level ($p = 1.8 \times 10^{-8}$), top-decile versus bottom-decile response 1.92 vs 0.89 (MWU 1.3×10^{-7}). Baseline reproduced to the digit ($r = +0.3739$, $n = 433$); metric invariance $\rho(\kappa^\mu, \kappa_V^{\text{lex}}) = 0.99998$ — both metrics sample the same flux-event structure along the same lexicographic trajectory, so the association is an event-structure property of the flux layer, not a metric artifact and not a value-function shadow. Cross-platform replication with the locked metric: the carbon-switch arms replicate on the same platform (M3D microarray: glycerol +0.195, acetate +0.166, proline +0.175, switch-MAX +0.206; $n = 424$, all $p < 6 \times 10^{-4}$) but dissociate across platforms — the PRECISE RNA-seq carbon-switch statistic (ten WT non-glucose carbon conditions) gives $r = -0.044$ (partial -0.088 , NS), with individual conditions weakly positive (glycerol +0.126, acetate +0.130,

fructose +0.099) except galactose (−0.088). The dissociation previously established with the precursor κ_V predictor is metric-robust: the association is specific to the carbon-exhaustion class and the platform-matched design, reported as an honest negative.

6.4 [E-V6] The layer decision

All value-layer attribution arms, on the frozen V5 protocol (iJO1366, E22 physiology, $n = 424$): the flux layer κ^μ gives $r = +0.395$, partial $r = +0.269$ ($p = 1.8 \times 10^{-8}$); the c -attribution arm is supported on the biomass pseudo-reaction alone (0 of 433 genes; Theorem 3.11(ii)); the shadow-price-jump arm gives 51 genes at $r = +0.032$ (partial -0.013 , $p = 0.93$). The value trajectory has *zero* chamber crossings: the path stays in one chamber of Φ throughout — Φ is affine along it, with $d\mu/dq_{\text{glc}} = Y = 0.099544$ ($= |y_{\text{glc}}|$, the constant glucose uptake shadow price; affine intercept -0.0124) and uptake shadow prices jumping by exactly zero — all four value kinks are anchor corners of the trajectory design, of magnitude $Y \Delta(dq_{\text{glc}}/dt)$, carrying no network structure. Value/flux strain mass ratio 1.45×10^{-3} ; the coupling identity holds at machine precision (error 0.0 at every kink). The value-gated flux strain coincides with κ^μ exactly (433/433 genes attain their maximum at a value-kink time), a property of the trajectory design (anchor rerouting dominates per-gene maxima), not evidence of value-layer content. The layer decision: the association lives on the flux layer, with Theorem 3.11 as the exact bridge to the canonical object.

6.5 [E-V7] Path robustness: what survives a trajectory change

Three trajectories through the same LP (frozen engine, seed, panel, statistics; only the path and its matched response vary): P0 glucose decline (E22 anchors; the V6 control), P1 oxygen limitation ($q_{\text{glc}} = 5$ fixed, q_{O_2} linear from 22 to 1), P2 acetate switch ($q_{O_2} = 22$ fixed; glucose $5 \rightarrow 0$ while acetate uptake ramps $0 \rightarrow 10$). Kinks are classified by the chamber test — the parameters are the uptake bounds, so $\nabla\Phi = (y_{\text{glc}}, y_{O_2}, y_{\text{ac}})$: a kink is a *design corner* iff it sits at a trajectory anchor and the uptake shadow prices are continuous across it (jump $\leq 10^{-9}$), and a *chamber crossing* iff they jump.

- (i) The one-chamber property is *path-dependent*: P0 has zero chamber crossings (V6 reproduced digit-exactly), while P1 has three genuine crossings ($q_{O_2} \approx 9.3, 6.3, 2.9$; $|\Delta\Phi'|$ up to 0.68; y_{O_2} jumps +0.032 at the respiratory boundary) and P2 has four (two large, near the glucose–acetate handoff; two micro-slivers with dual jumps $\sim 10^{-4}$). The E24 single-chamber structure is a property of the glucose-decline design, not a law.
- (ii) The layer decision survives the change: on every path the value-layer arms contribute nothing. The c -attribution arm is degenerate (0 of 433 genes — the sparse-objective corollary of Theorem 3.11 is path-free); the chamber-crossing shadow-price arm is null ($r = -0.17$, $n = 68$ on P1; $r = +0.03$, $n = 71$ on P2); value-gated flux strain is a subset of κ^μ (equal or weaker on every path, never stronger).
- (iii) The flux-layer association generalizes: against the matched oxygen response (M3D WT anaerobic vs aerobic contrast) P1 gives $r = +0.318$ (partial +0.258, $p = 6 \times 10^{-8}$, $n = 426$); against the matched acetate-switch response P2 gives $r = +0.223$ (partial +0.160, $p = 9 \times 10^{-4}$); and the P1/P2 predictors correlate with the E24 carbon response at $r = +0.378 / +0.391$ — the per-gene ranking is largely trajectory-independent.
- (iv) Theorem 3.11 holds at every kink on all three paths (identity error 0; worst case 1.6×10^{-15}). Value/flux strain mass ratios: 1.5×10^{-3} (P0), 2.4×10^{-4} (P1), 5.5×10^{-4} (P2).

6.6 [E-E25] Platform-by-class disambiguation (2×2)

Is the association a carbon-depletion artifact, or a platform artifact? Four cells (two platforms $\times$ two perturbation classes), executed with the precursor κ_V metric (rank-equivalent to κ^μ at $\rho = 0.99998$): carbon switch on microarray (M3D glycerol/acetate/proline vs glucose) switch-MAX $r = +0.196$; stress on microarray (heat shock, acid shock, ciprofloxacin) MAX $r = +0.298$; carbon starvation on RNA-seq (GSE64021 time course) MAX $r = +0.191$; heat shock on RNA-seq MAX $r = +0.345$ ($n = 241\text{--}433$; permutation $p \leq 0.003$). Signed responses flip sign (stress arms -0.22 to -0.32): the metric tracks *rerouting magnitude*, not direction, on both platforms and both classes — the association is neither platform-specific nor carbon-class-specific.

6.7 [E-E26] Protein abundance versus protein change

Two dissociations at the protein layer (PaxDb integrated abundance; GSE64021 quantitative proteomics; $n = 169\text{--}429$): (i) abundance *level* associates (PaxDb baseline $r = +0.334$) — high-rerouting genes are abundant proteins; (ii) abundance *change* does not (protein log-fold change $r = +0.008$ to $+0.032$; transcript–protein fold-change coupling $r = +0.020$, $n = 799$). The transcriptional association stops at translation, consistent with buffering.

6.8 [E-E27] Protein-layer null

Condition-dependent quantitative proteomics (22 conditions, triplicates, $n = 366$): protein-level correlation $r = -0.083$ against transcript $+0.419$ on the same genes. The transcript-level association does not propagate to the protein layer; buffering at translation is the consistent model, positioning Kochanowski et al. as prior art for the protein-layer null.

7 Discussion: one measure, several resolutions

The three sensitivity objects of the earlier literature — the geometric κ_V , the flux-statistical κ_{flux} , and the time-course κ_{time} — are one measure at different resolutions: the atomic measure of Definition 3.1, its coarse-grainings $\mu * \phi_\sigma$, and its integrals along parameter trajectories (κ^μ). The smooth member exists only through the window $h \ll \sigma \ll L_{\text{var}}$; the $\sigma \rightarrow 0$ limit is the atomic object. Open items, demoted to their honest status: Conjecture 4.6 (general complex stability), E28 under the (ε, σ) design law, and E32 (event-measure stabilization — the L^1 -reconstruction regime of Theorem 4.1(iv)). The formal identity $\kappa_{\text{flux}} = \mathcal{F}[\mu]$ is closed by Theorem 3.11 (the exact objective contraction $D^2\Phi = \sum_r c_r D^2v_r^*$), which supersedes the earlier metric-invariance argument for why the association stands on the flux layer.

Limitations. Correlational design; single-organism scope; operational (not combinatorial) active sets; lexicographic tie-breaking is an engine convention (the value-function carrier is immune, and the flux layer declares it — Remark 3.5); the one-chamber structure of the E24 path is condition-specific (E-V7) while the layer decision itself is path-robust; the cross-platform carbon-switch dissociation (PRECISE, E-V5) bounds the generality of the association; categorical and homotopy-theoretic material of the predecessor manuscript is retained only where it carries falsifiable content, with the remainder moved to a companion document.

What this paper is and is not. The mathematics is classical: parametric linear programming (chamber complexes, active sets, Danskin duality), Alexandrov’s theorem, and distributional second derivatives of piecewise-affine maps; Theorem 3.11 is an elementary identity. The contribution is the systematic *application* of this machinery to flux-balance models — the two-layer measure structure of the value/flux pair, the exact coupling identity, the signed extension to arbitrary GPR structure — together with the empirical association and its layer decision. No

new measure-theoretic theorem is claimed; the nearest prior art (phenotype phase planes, mpLP value-function theory, discrete Monge–Ampère, Regge-type defect measures) is cited where it applies.

8 Methods

Engine. Three-stage lexicographic pFBA on HiGHS (lexicographic order: biomass, then ℓ_1 flux, then seeded deterministic weights); vertex degeneracy of plain pFBA documented (0.69 flux flips between warm-started solves) and resolved deterministically.

Data. M3D microarray panels (carbon-depletion contrast; WT anaerobic-vs-aerobic contrast; MOPS carbon-source switches); PRECISE 2.0 evaluated (no carbon-depletion class, documented); PRECISE WT carbon-switch conditions used for the cross-platform arm; Schmidt et al. 2016 condition-dependent proteome; PaxDb integrated abundance. SHA-256 provenance manifests deposited.

Reproducibility. All experiments deposit artifacts (npz/csv/json) with scripts; independent-seed reproductions recorded for the refinement prototype and the stress batteries. The V6/V7 layer decisions and the PRECISE arm re-run in under two minutes each (`e24_layer_decision.py`, `v7_path_robustness.py`, `precise_arm_kappamu.py`).

A Layer-1 porting map

v2 section	v21 source (frozen; do not modify)
§1	v21 sec 1 (rewritten to the single story)
§2	v21 sec 2 (subset: active-set notation only)
§3	v21 active-set stratification defs; new D1/D2
§4	new (Theorem B'; replaces the smooth bridge)
§5	M1/M3/M4 report bodies
§6	v21 E22–E27 sections (E22, E23, E25, E26 ported; E24 $\rightarrow$ E-V5 with the PRECISE arm; E27 $\rightarrow$ E-E27; V6/V7 new)
§7	v21 discussion, rewritten as resolution statement
§8	v21 methods; engine documentation
companion	categorical/HoTT sections of v21 (trimmed subset)